UNIT 1 FUNCTIONS, LIMITS AND CONTINUITY

1.1 Functions

- **Definition**

> A **function** f is a rule that assigns to each element x in a set D exactly one element, called $f(x)$, in a set E.

- D is called the **domain** of the function, and E is called the **codomain**.
- The **range** of f is the set of all possible values of $f(x)$ as x ranges throughout the domain.
- A symbol that represents an arbitrary number in the domain of a function f is called an **independent variable**.
- A symbol that represents a number in the range of f is called the **dependent variable**.
- **Example** Given a function $y = f(x)$ where $f(0) = 4$, $f(1) = -1$ and $f(2) = 0$.

  Domain, $D =$

  Codomain, $E =$

  Range, $R =$

  Independent Variable $=$

  Dependent Variable $=$

- Examples of functions:

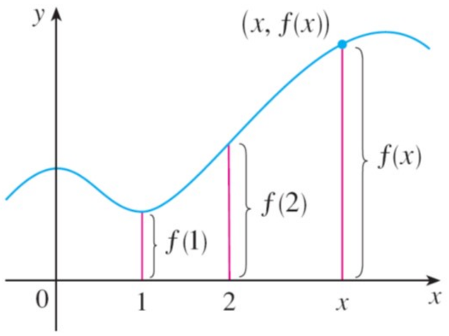
  (a) The area of a circle depends on the radius, $A = \pi r^2$.

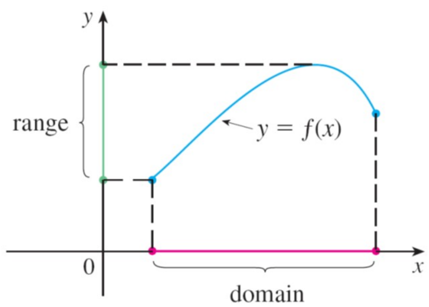
  (b) The world population P depends on time t.

  (c) Your final mark y depends on the amount of time you spend on your studies x.

- The graph of a function with domain D is the set of ordered pairs

$$\{(x, f(x)) \,|\, x \in D\}$$

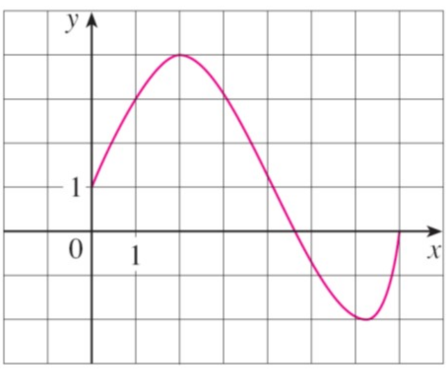




- **Example** The graph of a function f is shown below.

  (a) Find the values of $f(1)$ and $f(5)$.

  (b) What are the domain and range of f?



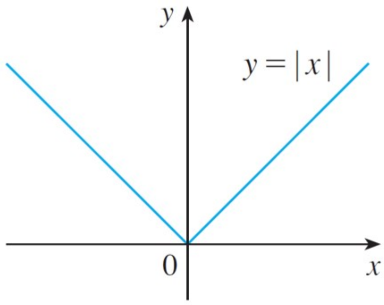
- **Piecewise Defined Function**
- **Example** A function f is defined by

$$f(x) = \begin{cases} 1 - x & \text{if } x \leq -1 \\ x^2 & \text{if } x > -1 \end{cases}$$

Evaluate $f(-2)$, $f(-1)$ and $f(0)$, then sketch the graph.

- **Absolute Values**

$$|x| = \begin{cases} x, & \text{if } x \geq 0 \\ -x, & \text{if } x < 0 \end{cases}.$$



- **Properties of Absolute Values**

1. $|ab| = |a|\,|b|$.
2. $\left|\dfrac{a}{b}\right| = \dfrac{|a|}{|b|}$.
3. $|a+b| \leq |a| + |b|$.
4. $|a-b| \geq ||a| - |b||$.

- $|x| < a$ if and only if $-a < x < a$.

  $|x| > a$ if and only if $x < -a$ or $x > a$.

- **Example** Evaluate the following:

  $|3| =$

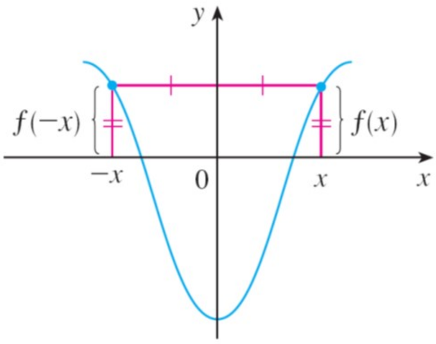
  $|-3| =$

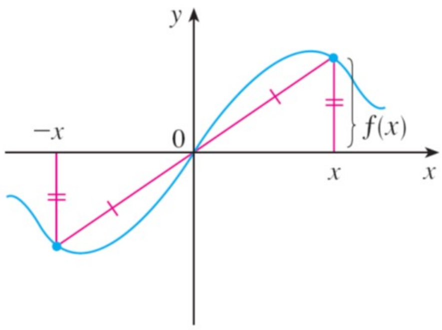
  $|\sqrt{2} - 1| =$

  $|3 - \pi| =$

- **Symmetry**
- A function $y = f(x)$ is **even** if and only if $f(-x) = f(x)$ for all x. The graphs of even functions are symmetric with respect to the y-axis.

- A function $y = f(x)$ is **odd** if and only if $f(-x) = -f(x)$ for all x. The graphs of odd functions are symmetric with respect to the origin.





- **Example** Determine whether each of the following functions is odd, even or neither.

  (a) $f(x) = x^5 + x$

  (b) $f(x) = 1 - x^4$

  (c) $f(x) = 2x - x^2$

- **Increasing and Decreasing**

Let f be defined on an interval I. We say that

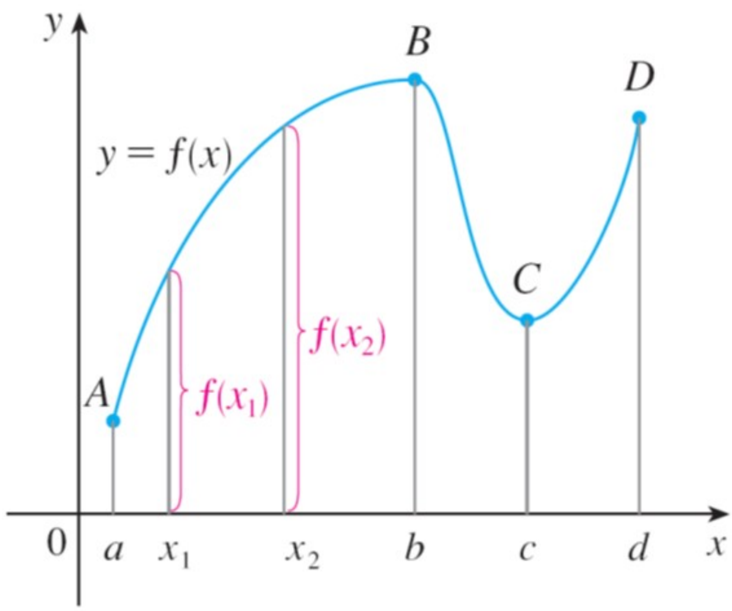
(i) f is **increasing** on I if, for every pair of numbers x_1 and x_2 in I,

$$x_1 < x_2 \quad \Longrightarrow \quad f(x_1) < f(x_2)$$

(ii) f is **decreasing** on I if, for every pair of numbers x_1 and x_2 in I,

$$x_1 < x_2 \quad \Longrightarrow \quad f(x_1) > f(x_2)$$

(iii) f is strictly monotonic on I if it is either increasing on I or decreasing on I.



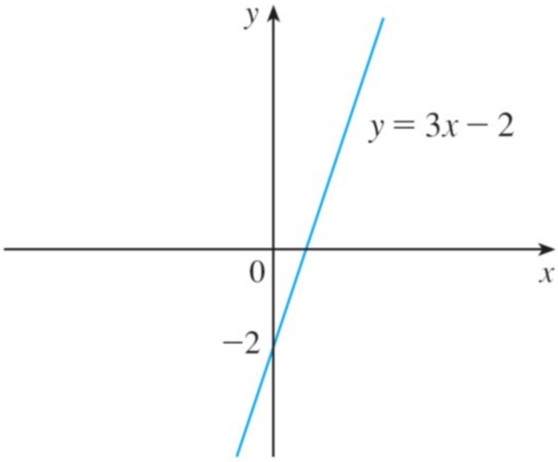
Exercises 1.1 Page 19: 1, 2 (no, why?), 3, 7, 9, 27, 29, 31, 35, 37, 43, 45, 59, 63, 71, 73, 75, 77, 79.

1.2 A Catalog of Essential Functions

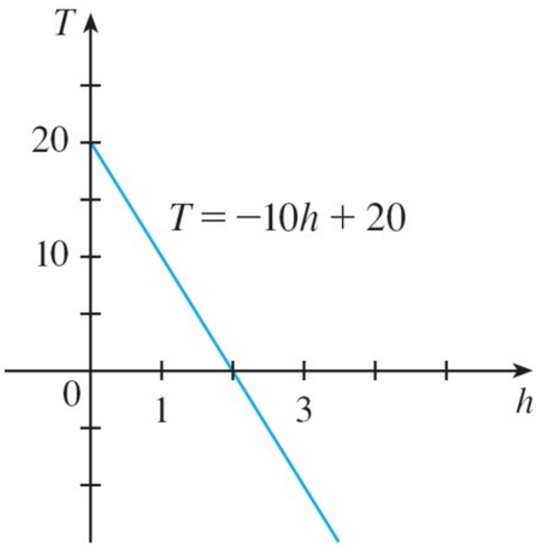
- **Linear Models**

$$y = mx + b$$

- Positive Linear



- Negative Linear

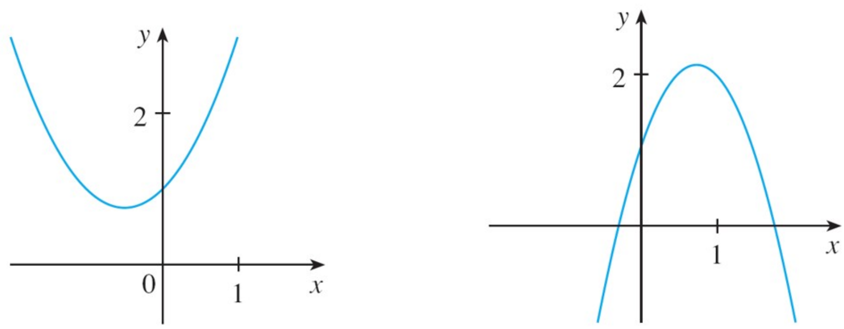


- **Polynomials** A function P is called a polynomial if

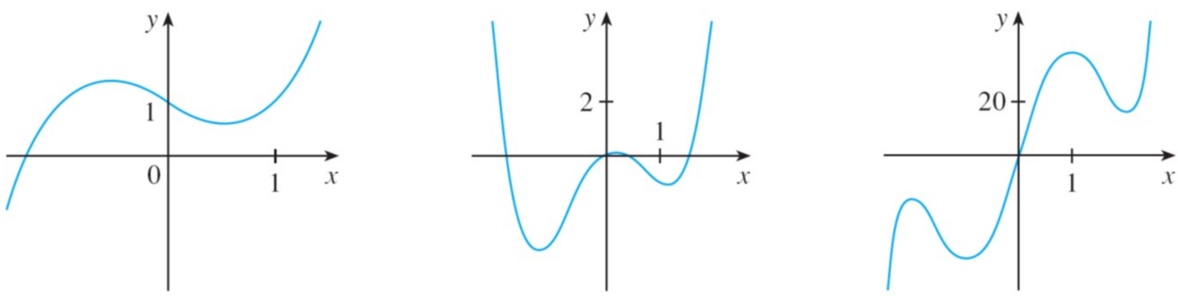
$$P(x) = a_n x^n + a_{n-1} x^{n-1} + \ldots + a_2 x^2 + a_1 x + a_0 \qquad a_n \neq 0$$

a_0, …, a_n are constants (*coefficients*). a_n is called the leading coefficient, and the polynomial is of degree n.

- If $n = 1$, it is called a **linear** function.
- If $n = 2$, it is called a **quadratic** function.
- If $n = 3$, it is called a **cubic** function.

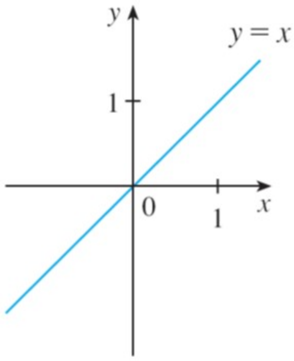


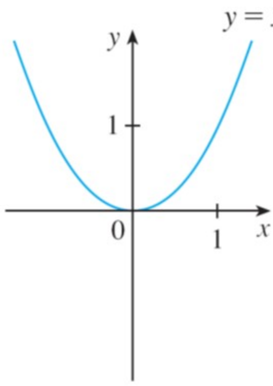
(a) $y = x^2 + x + 1$ (b) $y = -2x^2 + 3x + 1$

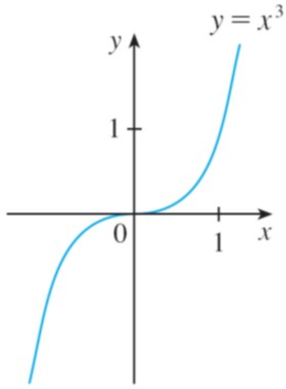


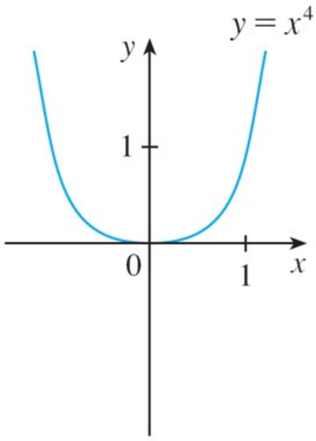
(a) $y = x^3 - x + 1$ (b) $y = x^4 - 3x^2 + x$ (c) $y = 3x^5 - 25x^3 + 60x$

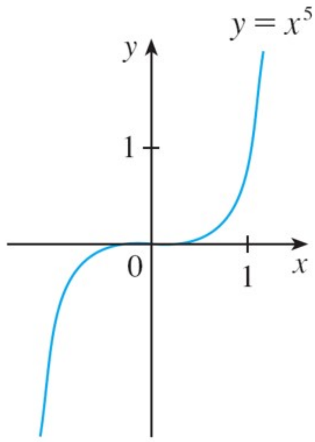
- **Power Functions** A function of the form $f(x) = x^{\alpha}$, where α is a constant, is called a power function.
  - When $\alpha = n$ is an integer, they are polynomial functions.
  - When $\alpha = \dfrac{1}{n}$, where n is a positive integer, they are called **root** functions.
  - When $\alpha = -1$, it is called a **reciprocal** function.

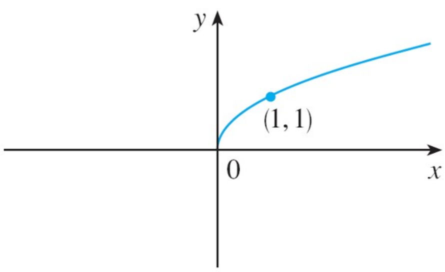




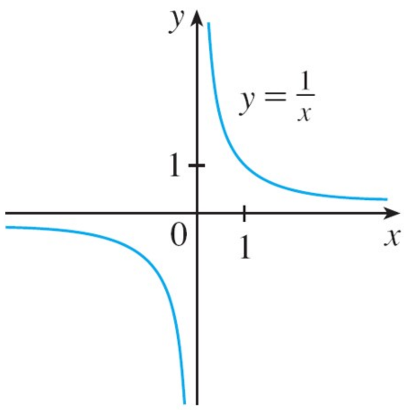








$f(x) = \sqrt{x}$

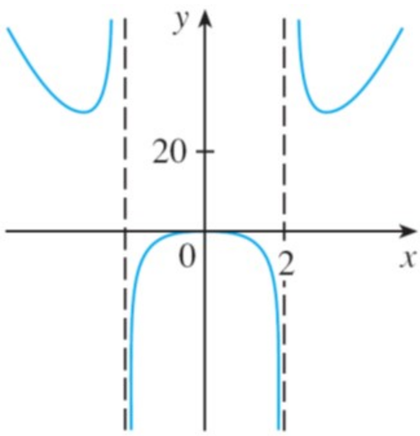


- **Rational Functions** Rational function is a function that can be written as the ratio of two polynomials:

$$f(x) = \frac{P(x)}{Q(x)}$$

An example of a rational function is

$$f(x) = \frac{2x^4 - x^2 + 1}{x^2 - 4}$$



- **Algebraic Functions** An explicit algebraic function is a function that can be obtained from polynomials via addition, subtraction, multiplication, division and root extraction. Eg,

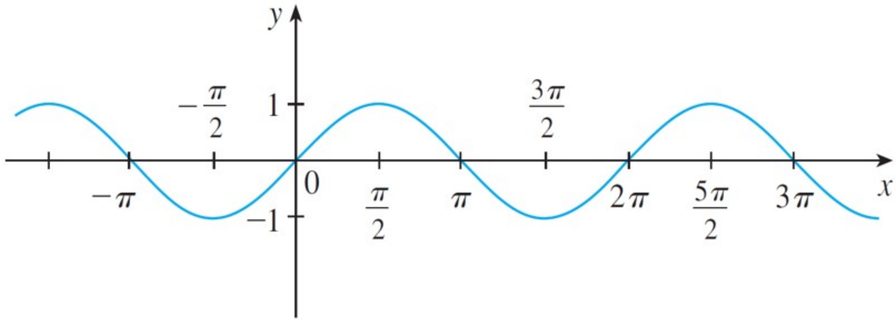
(a) $f(x) = 3x^{\frac{2}{5}}$

(b) $g(x) = \dfrac{(x+2)\sqrt{x}}{x^3 + \sqrt[3]{x^2 - 1}}$

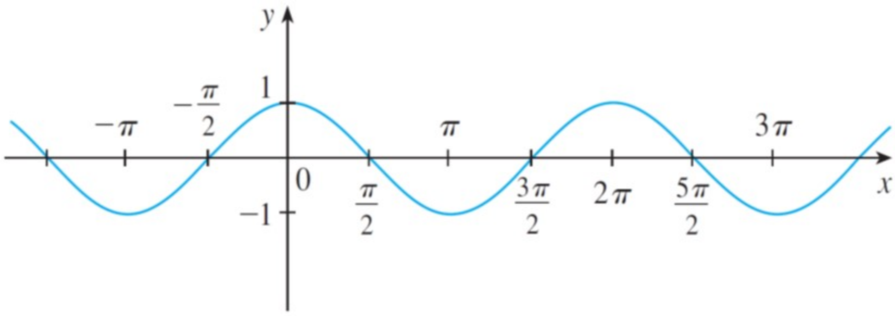
(c) $h(x) = \dfrac{x^4 - 16x^2}{x + \sqrt{x}} + (x-2)\sqrt[3]{x+1}$

- **Trigonometric Functions** In calculus, radian measure is always used except when otherwise specified.

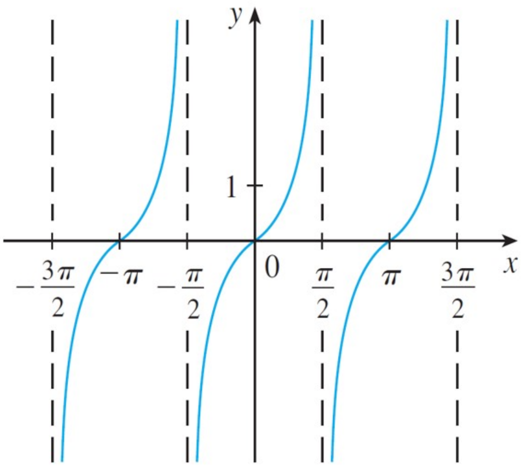
  - $\sin x$



  - $\cos x$



  - $\tan x = \dfrac{\sin x}{\cos x}$



  - $\cot x = \dfrac{\cos x}{\sin x}$
  - $\sec x = \dfrac{1}{\cos x}$
  - $\csc x = \dfrac{1}{\sin x}$

- $-1 \leq \sin x \leq 1$

- $-1 \leq \cos x \leq 1$

- $y = \sin x$ and $y = \tan x$ are odd functions.

- $y = \cos x$ is an even function.

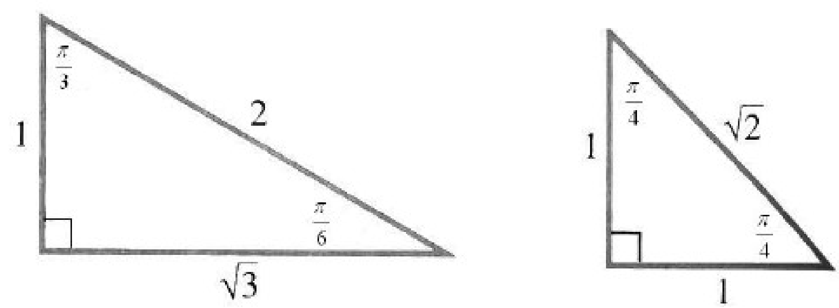
- A function is periodic if there is a positive number p such that

$$f(x + p) = f(x)$$

  for all real numbers x in the domain of f. The smallest such p is called a **period** of f.

  - $\sin(x + 2\pi) = \sin x$
  - $\cos(x + 2\pi) = \cos x$
  - $\tan(x + \pi) = \tan x$

- Special right-angled triangles:



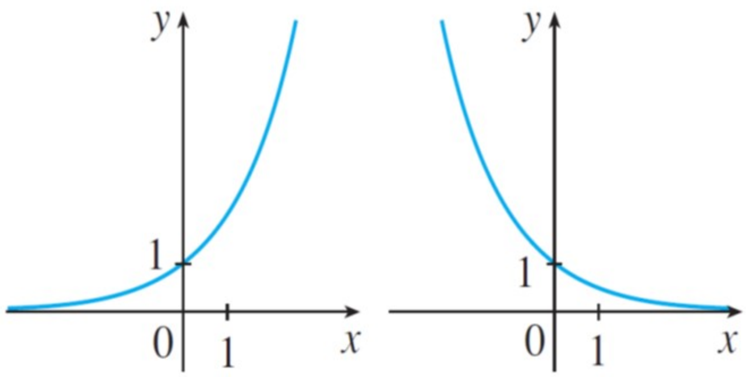
- **Trigonometric Identities**

1. $\sin^2\theta + \cos^2\theta = 1$
2. $1 + \tan^2\theta = \sec^2\theta$
3. $1 + \cot^2\theta = \csc^2\theta$
4. $\sin(A \pm B) = \sin A \cos B \pm \cos A \sin B$
5. $\cos(A \pm B) = \cos A \cos B \mp \sin A \sin B$
6. $\tan(A \pm B) = \dfrac{\tan A \pm \tan B}{1 \mp \tan A \tan B}$
7. $\sin 2\theta = 2\sin\theta\cos\theta$
8. $\cos 2\theta = \cos^2\theta - \sin^2\theta = 2\cos^2\theta - 1 = 1 - 2\sin^2\theta$
9. $\tan 2\theta = \dfrac{2\tan\theta}{1 - \tan^2\theta}$
10. $\sin^2\theta = \dfrac{1 - \cos 2\theta}{2}$
11. $\cos^2\theta = \dfrac{1 + \cos 2\theta}{2}$
12. $\sin A + \sin B = 2\sin\dfrac{A+B}{2}\cos\dfrac{A-B}{2}$
13. $\cos A + \cos B = 2\cos\dfrac{A+B}{2}\cos\dfrac{A-B}{2}$
14. $\sin A \sin B = \dfrac{\cos(A-B) - \cos(A+B)}{2}$
15. $\cos A \cos B = \dfrac{\cos(A+B) + \cos(A-B)}{2}$
16. $\sin A \cos B = \dfrac{\sin(A+B) + \sin(A-B)}{2}$

- **Exponential Functions** An exponential function is a function of the form

$$f(x) = b^x$$

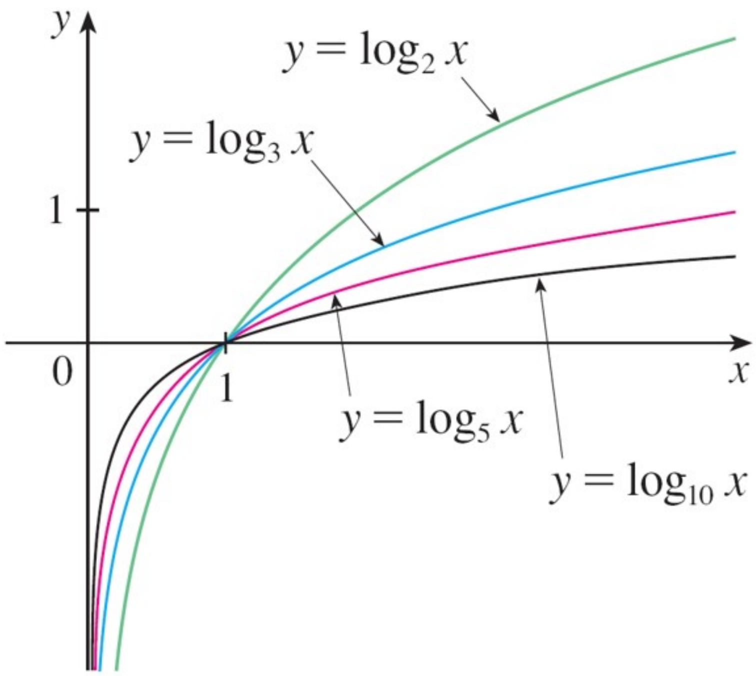
where the base b is a positive constant.



- **Logarithmic Functions** A logarithmic function is a function of the form

$$f(x) = \log_b x$$

where the base b is a positive constant.



Exercises 1.2 Page 33: 1, 3, 7, 11, 13, 19.

1.3 New Functions from Old Functions

- **Translations, Stretching and Reflecting**

- **Example** The graphs of the following functions are given below. (a) is the original function. Identify the rest as "translation", "stretching" or "reflecting".

  (a) $y = \sqrt{x}$ original

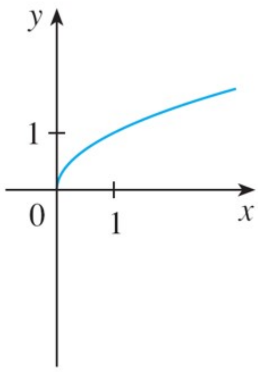
  (b) $y = \sqrt{x} - 2$

  (c) $y = \sqrt{x - 2}$

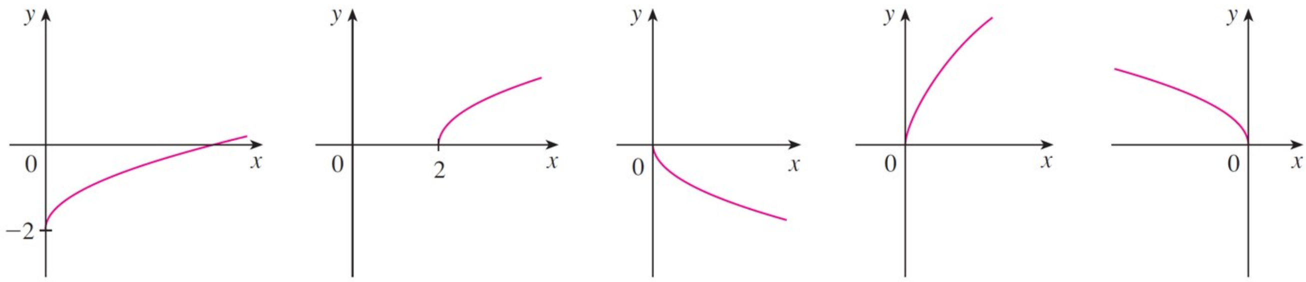
  (d) $y = -\sqrt{x}$

  (e) $y = 2\sqrt{x}$

  (f) $y = \sqrt{-x}$



(a) $y = \sqrt{x}$



(b) $y = \sqrt{x} - 2$ (c) $y = \sqrt{x - 2}$ (d) $y = -\sqrt{x}$ (e) $y = 2\sqrt{x}$ (f) $y = \sqrt{-x}$

- **Example** Sketch the graph of $f(x) = x^2 + 6x + 10$.

- **Example** Sketch the graph of $y = \sin 2x$ and $y = 1 - \sin x$.

- **Combinations of Functions**

- **Example** Given $f(x) = \dfrac{x-3}{2}$ and $g(x) = \sqrt{x}$, find

(a) $f + g$

(b) $f - g$

(c) $f \cdot g$

(d) $\dfrac{f}{g}$

- Given two functions f and g, the **composition** of g and f is the function $g \circ f$ such that

$$g \circ f(x) = g\left(f(x)\right)$$

- **Example** Given $f(x) = \sqrt{x}$ and $g(x) = \sqrt{2-x}$, find each of the following functions and their domains.

  (a) $g \circ f$

  (b) $f \circ g$

  (c) $f \circ f$

  (d) $g \circ g$

- **Example** Given $F(x) = \cos^2(x+9)$, find functions f, g and h such that $F = f \circ g \circ h$.

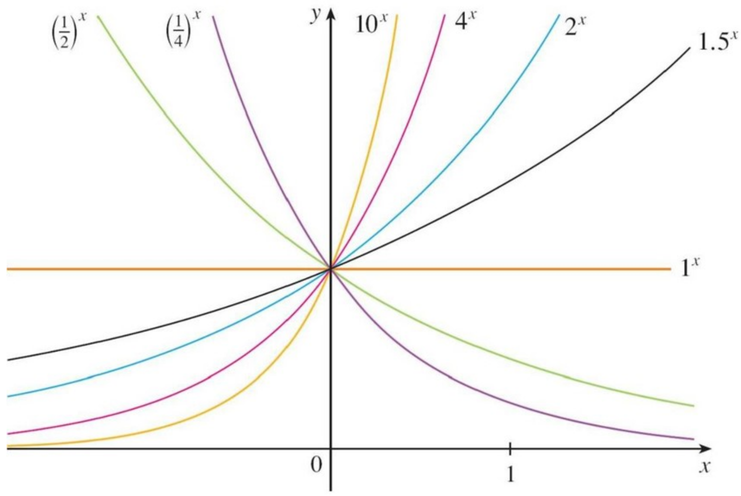
Exercises 1.3 Page 42: 1, 3, 7, 9, 11, 13, 15, 17, 19, 21, 23, 29, 31, 37, 41, 43, 45, 47, 49, 63.

1.4 Exponential Functions

- An exponential function is a function of the form

$$f(x) = b^x$$

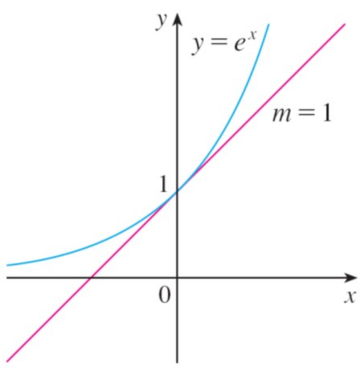
  where $b > 0$ is a constant.



- The number e is a number

$$e = 2.71828...$$

  It is the number such that the slope of the tangent line to the curve $y = e^x$ at $x = 0$ is exactly 1.



- $f(x) = e^x$ is called the **natural** exponential function.

- **Example** Graph the function $y = \dfrac{1}{2}e^{-x} - 1$ and state the domain and the range.

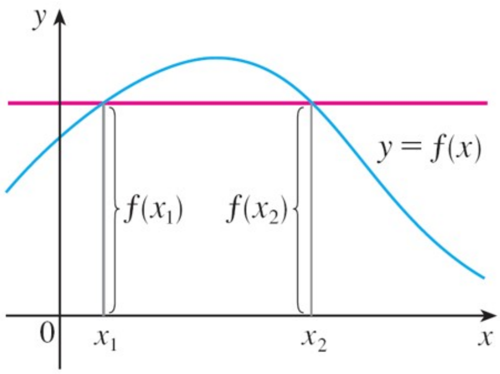
Exercises 1.4 7 Page 53: 11, 15 (use Figure 16), 17, 19.

1.5 Inverse Functions and Logarithms

- **One-to-One functions**

A function f is called a **one-to-one function** if it never takes on the same value twice, i.e.,

$$\text{if } x_1 \neq x_2 \quad \Longrightarrow \quad f(x_1) \neq f(x_2)$$



- **Example** Is the function $f(x) = x^3$ one-to-one?

- **Example** Is the function $f(x) = x^2$ one-to-one?

- **Inverse Function**

Let f be a one-to-one function with domain A and range B. Its **inverse function** f^{-1} has domain B and range A and is defined by

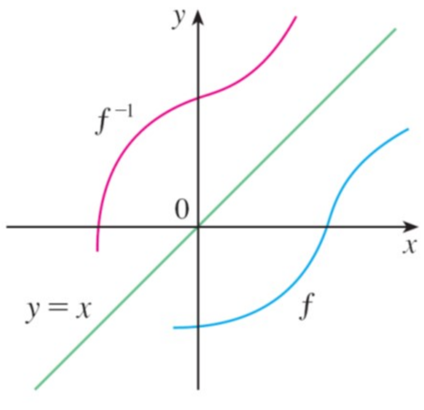
$$f^{-1}(y) = x \quad \Leftrightarrow \quad y = f(x)$$

-

$$f^{-1}\left(f(x)\right) = x \qquad\qquad f\left(f^{-1}(x)\right) = x$$

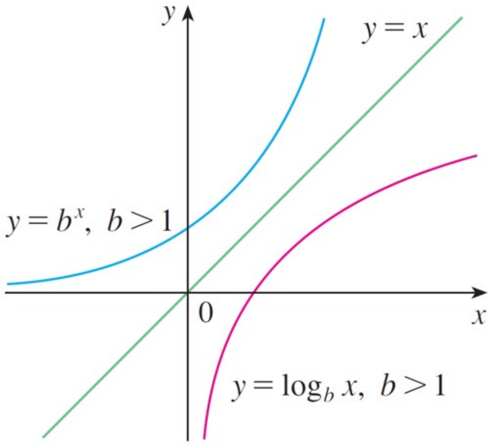
- **Example** Find the inverse function of $f(x) = x^3 + 2$.

The graph of f^{-1} is obtained by reflecting the graph of f about the line $y = x$.



- **Logarithmic Functions**

$$\log_b x = y \quad \Longleftrightarrow \quad x = b^y$$



-

$$\log_b(b^x) = x$$
$$b^{\log_b x} = x$$

- **Natural Logarithms**

$$\log_e x = \ln x$$
$$\ln x = y \quad \Longleftrightarrow \quad x = e^y$$
$$\ln(e^x) = x$$
$$e^{\ln x} = x$$
$$\ln e = 1$$

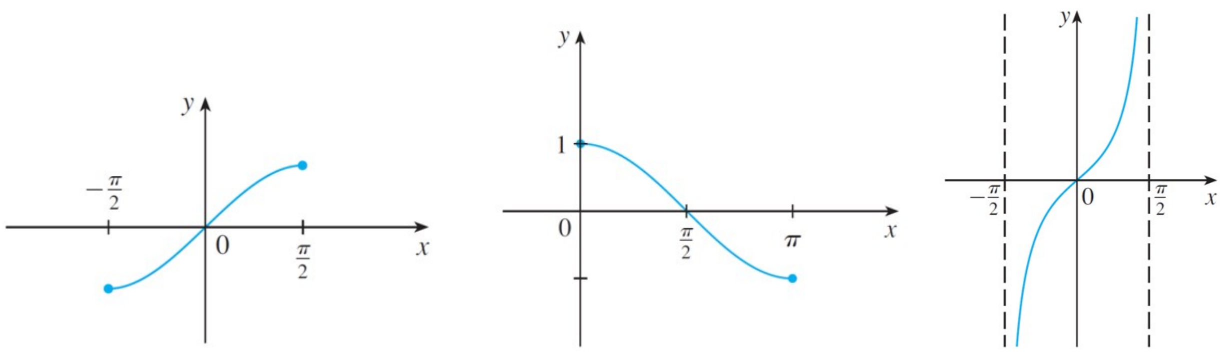
- **Example** Sketch the graph of the function $y = \ln(x-2) - 1$.

- **Inverse Trigonometric Functions**

$$x = \sin^{-1} y \iff y = \sin x, \qquad -\frac{\pi}{2} \le x \le \frac{\pi}{2}$$

$$x = \cos^{-1} y \iff y = \cos x, \qquad 0 \le x \le \pi$$

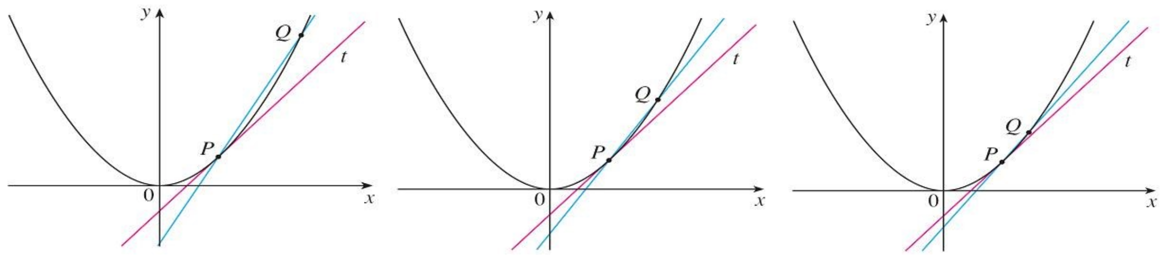
$$x = \tan^{-1} y \iff y = \tan x, \qquad -\frac{\pi}{2} \le x \le \frac{\pi}{2}$$

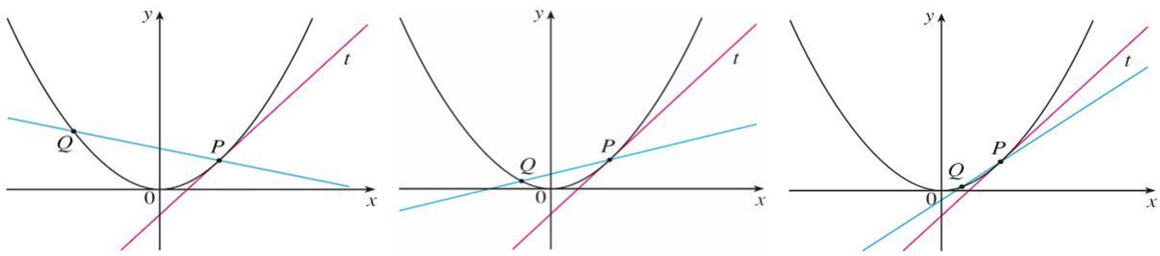


Exercises 1.5 Page 66: 5, 7, 9, 11, 15, 21, 23, 25, 27, 31, 47, 49, 75.

2.2 The Limit of a Function

- Consider a function $y = f(x)$. How are we going to find the equation of the tangent line to the curve at the point $P(a, f(a))$?

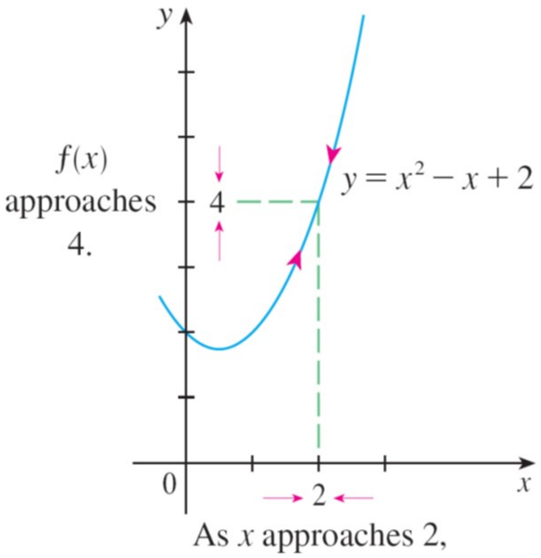




- How do you find the instantaneous velocity of a moving object?

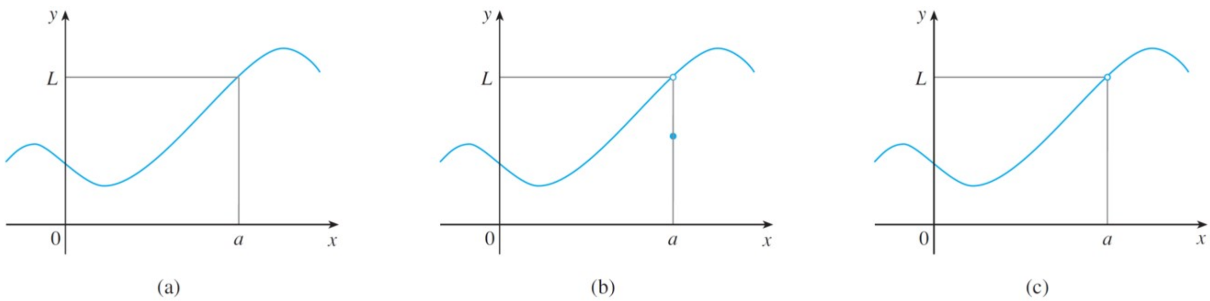
- Calculus is the study of limits.

- How do we find the limit of $x^2 - x + 2$ as x approaches 2?



- **Definition**

> Suppose that $f(x)$ is defined when x is near the number a. We write
>
> $$\lim_{x \to a} f(x) = L$$
>
> and say that
>
> the **limit** of $f(x)$, as x approaches a, equals L
>
> if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a (on either side of a) but not equal to a.



- **Definition**

We write

$$\lim_{x \to a^-} f(x) = L$$

and say that

the **left-hand** limit of $f(x)$, as x approaches a from the left, equals L

if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a with x less than a.

- **Definition**

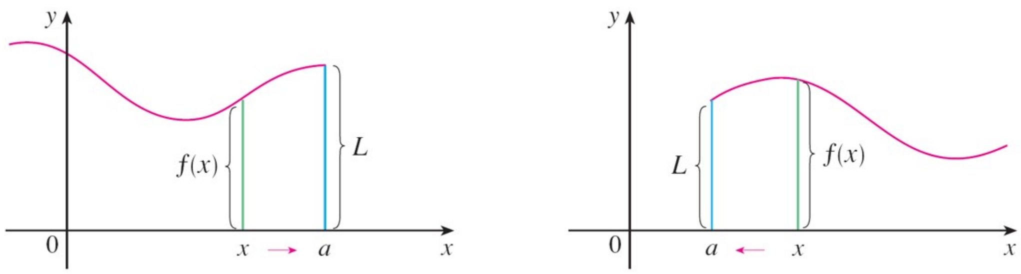
We write

$$\lim_{x \to a^+} f(x) = L$$

and say that

the **right-hand** limit of $f(x)$, as x approaches a from the right, equals L

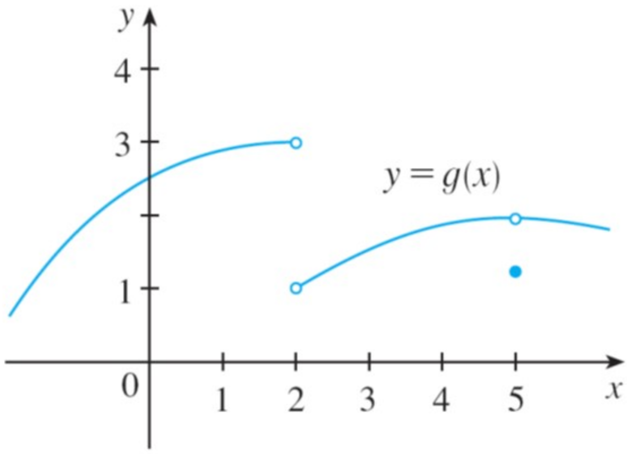
if we can make the values of $f(x)$ arbitrarily close to L by restricting x to be sufficiently close to a with x greater than a.



- **Theorem**

$\lim_{x \to a} f(x) = L$ if and only if $\lim_{x \to a^+} f(x) = L$ and $\lim_{x \to a^-} f(x) = L$.

- **Example** The graph of a function g is shown below. Use it to state the values (if they exist) of the following:



(a) $\lim_{x \to 2^-} g(x)$

(b) $\lim_{x \to 2^+} g(x)$

(c) $\lim_{x \to 2} g(x)$

(d) $\lim_{x \to 5^-} g(x)$

(e) $\lim_{x \to 5^+} g(x)$

(f) $\lim_{x \to 5} g(x)$

(g) $f(5)$

Infinite Limits

- **Example** Find $\lim_{x \to 0} \dfrac{1}{x^2}$ if it exists.

- **Intuitive Definition of an Infinite Limit**

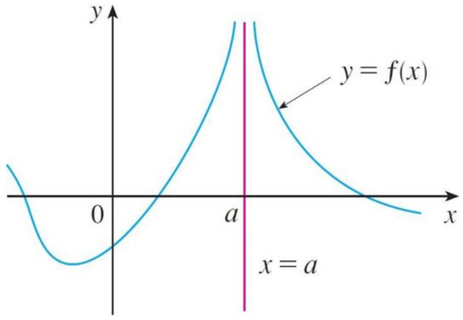
Let f be a function defined on both sides of a, except possibly at a itself. Then

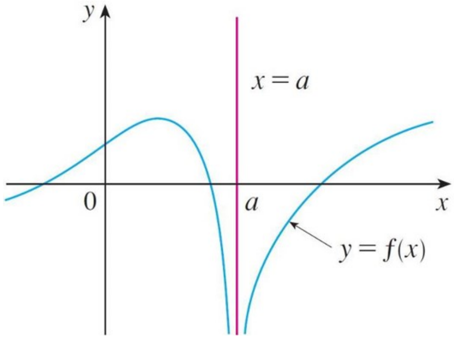
$$\lim_{x \to a} f(x) = \infty$$

means that the values of $f(x)$ can be made arbitrarily large by taking x sufficiently close to a, but not equal to a.

$$\lim_{x \to a} f(x) = -\infty$$

means that the values of $f(x)$ can be made arbitrarily large negative by taking x sufficiently close to a, but not equal to a.





- **Example** Find $\displaystyle\lim_{x \to 3^+} \frac{1}{x-3}$ and $\displaystyle\lim_{x \to 3^-} \frac{1}{x-3}$.

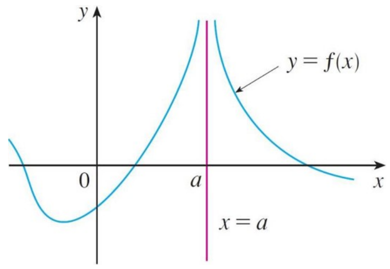
- **Example** Find $\displaystyle\lim_{x \to -3^+} \frac{2x}{x+3}$ and $\displaystyle\lim_{x \to -3^-} \frac{2x}{x+3}$.

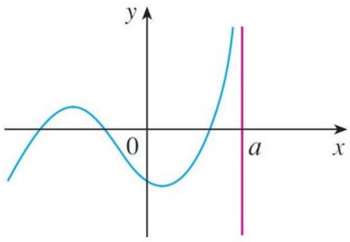
- **Vertical Asymptotes**

The vertical line $x = a$ is called a **vertical asymptote** of the curve $y = f(x)$ if at least one of the following statements is true.

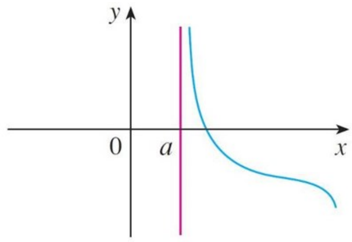
$$\lim_{x \to a} f(x) = \infty \qquad \lim_{x \to a^-} f(x) = \infty \qquad \lim_{x \to a^+} f(x) = \infty$$

$$\lim_{x \to a} f(x) = -\infty \qquad \lim_{x \to a^-} f(x) = -\infty \qquad \lim_{x \to a^+} f(x) = -\infty$$

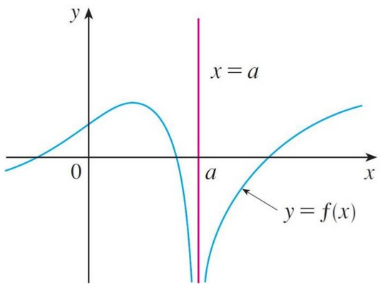


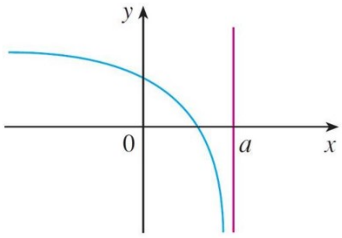


(a) $\lim_{x \to a^-} f(x) = \infty$

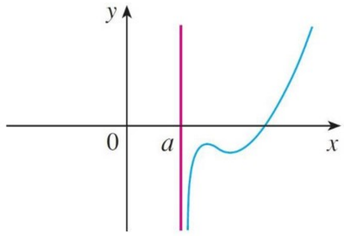


(b) $\lim_{x \to a^+} f(x) = \infty$





(c) $\lim_{x \to a^-} f(x) = -\infty$



(d) $\lim_{x \to a^+} f(x) = -\infty$

Exercises 2.2 Page 92: 5, 7, 9, 11, 19, 23, 27, 31, 33, 35, 41, 43, 45(a) + find the vertical asypmtote of the function.

2.3 Calculating Limits Using the Limit Laws

- **Limit Laws**

Let c be a constant, and f and g be functions that have limits at a. Then

1. $\lim\limits_{x\to a} cf(x) = c\lim\limits_{x\to a} f(x)$.
2. $\lim\limits_{x\to a} [f(x)+g(x)] = \lim\limits_{x\to a} f(x) + \lim\limits_{x\to a} g(x)$.
3. $\lim\limits_{x\to a} [f(x)-g(x)] = \lim\limits_{x\to a} f(x) - \lim\limits_{x\to a} g(x)$.
4. $\lim\limits_{x\to a} [f(x)\cdot g(x)] = \lim\limits_{x\to a} f(x) \cdot \lim\limits_{x\to a} g(x)$.
5. $\lim\limits_{x\to a} \dfrac{f(x)}{g(x)} = \dfrac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$ provided that $\lim\limits_{x\to a} g(x) \neq 0$.
6. $\lim\limits_{x\to a} c = c$.
7. $\lim\limits_{x\to a} x = a$.
8. $\lim\limits_{x\to a} [f(x)]^n = \left[\lim\limits_{x\to a} f(x)\right]^n$, n is a positive integer.
9. $\lim\limits_{x\to a} \sqrt[n]{f(x)} = \sqrt[n]{\lim\limits_{x\to a} f(x)}$.

- **Example** Find $\lim\limits_{x\to 3} 2x^4$.

- **Example** Find $\lim_{x \to 4} \left(3x^2 - 2x\right)$.

- **Example** Find $\lim_{x \to 4} \dfrac{\sqrt{x^2+9}}{x}$.

- **Example** If $\lim_{x \to 3} f(x) = 4$ and $\lim_{x \to 3} g(x) = 8$, find $\lim_{x \to 3} \left[f^2(x) \cdot \sqrt[3]{g(x)}\right]$.

- **Substitution Property**

> If f is a **polynomial function** or a **rational function**, then
>
> $$\lim_{x \to a} f(x) = f(a)$$
>
> provided that $f(a)$ is defined.

- **Example** Find $\lim_{x\to 2} \dfrac{7x^5 - 10x^4 - 13x + 6}{3x^2 - 6x - 8}$.

- **Example** Find $\lim_{x\to 1} \dfrac{x^3 + 3x + 7}{x^2 - 2x + 1}$.

- **Theorem**

> Suppose $f(x) = g(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, and $\lim_{x\to a} g(x)$ exists, then $\lim_{x\to a} f(x)$ exists and $\lim_{x\to a} f(x) = \lim_{x\to a} g(x)$.

- **Example** Find $\lim\limits_{x\to 1} \dfrac{x^2-1}{x-1}$.

- **Example** Find $\lim\limits_{x\to 2} \dfrac{x^2+3x-10}{x^2+x-6}$.

- **Example** Given $g(x) = \begin{cases} \dfrac{x-2}{x^2-4}, & x \neq 2 \\ 6, & x = 2 \end{cases}$, find $\lim\limits_{x\to 2} g(x)$.

- **Example** Find $\lim\limits_{x\to 1} \dfrac{x-1}{\sqrt{x}-1}$.

- **Example** Evaluate $\lim_{x\to 0} \dfrac{\sqrt{x^2+9}-3}{x^2}$.

- Some limits are best calculated by first finding the **left-** and **right-** hand limits.

- **Example** Find the limit $\lim_{x\to 0} |x|$.

- **Example** Does the limit $\lim_{x\to 0} \dfrac{|x|}{x}$ exists?

- **Example** Given $f(x) = \begin{cases} \sqrt{x-2}, & x > 6 \\ x-4, & x < 6 \end{cases}$, find $\lim_{x\to 6} f(x)$.

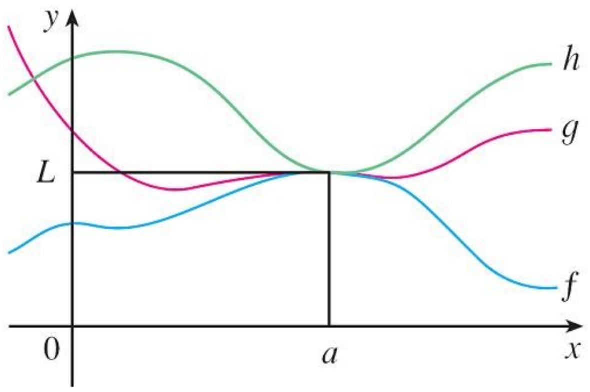
- **Squeeze Theorem/Sandwich Theorem**

Suppose $f(x) \leq g(x) \leq h(x)$ for all x in an open interval containing the number a, except possibly at the number a itself, and

$$\lim_{x \to a} f(x) = \lim_{x \to a} h(x) = L$$

then

$$\lim_{x \to a} g(x) = L$$



- **Example** Find $\lim_{x \to 0} x^2 \sin\left(\frac{1}{x}\right)$.

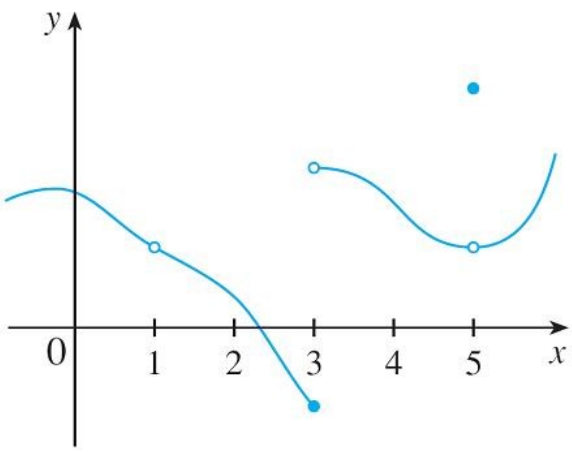
Exercises 2.3 Page 102: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21, 23, 25, 27, 29, 31, 37, 39, 41, 43, 47, 51, 53, 59, 61 (*Hint*: $0 \leq f(x) \leq x^2$ for all x.).

2.5 Continuity of Functions

- **Definition**

We say that f is **continuous** at a number a if

$$\lim_{x \to a} f(x) = f(a)$$



- **Example** Determine whether the function $f(x) = \dfrac{x^2-4}{x-2}$ is continuous at $x = 2$.

- **Example** Determine whether the function $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ 2, & x = 2 \end{cases}$ is continuous at $x = 2$.

- **Example** Let $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ c, & x = 2 \end{cases}$. Find the value of c such that f is continuous at $x = 2$.

- **Example** Determine all points of discontinuity of $f(x) = \dfrac{\sin x}{x(1-x)}$.

- **Continuity on an Interval**

> We say that f is continuous on an open interval if it is continuous at each point of that interval. It is **continuous** on the closed interval $[a, b]$ if it is continuous on (a, b), **right continuous** at a and **left continuous** at b.

- **Example** Show that the function $f(x) = 1 - \sqrt{1 - x^2}$ is continuous on the interval $[-1, 1]$.

- **Theorem**

If f and g are continuous at a, and k is a constant, then the following functions are also continuous at a:

- kf
- $f+g$
- $f-g$
- $f \cdot g$
- $\dfrac{f}{g}$, provided that $g(a) \neq 0$
- f^n
- $\sqrt[n]{f}$, provided that $f(a) > 0$.

- **Examples of Continuous Functions**

7 Theorem The following types of functions are continuous at every number in their domains:

- polynomials
- rational functions
- root functions
- trigonometric functions
- inverse trigonometric functions
- exponential functions
- logarithmic functions

- **Example** Use continuity to evaluate the limit.

(a) $\lim_{x \to \pi} \sin x$

(b) $\lim_{x \to 4} 3^x$

- **Composite Limit Theorem**

If $\lim_{x\to a} g(x) = L$ and f is continuous at L, then

$$\lim_{x\to a} f\left(g(x)\right) = f\left(\lim_{x\to a} g(x)\right) = f(L)$$

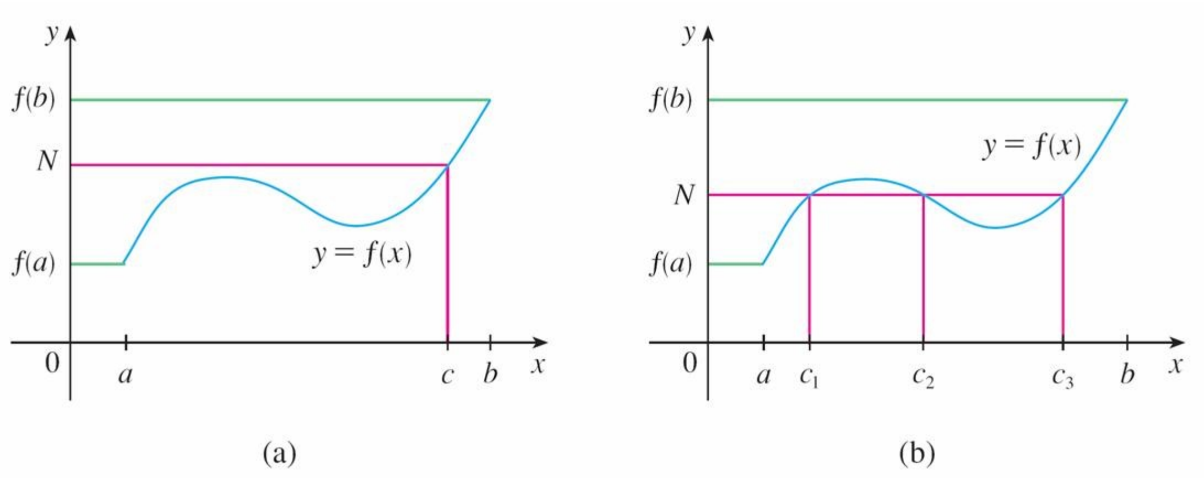
In particular, if g is continuous at a and f is continuous at $g(a)$, then the composite $f \circ g$ is continuous at a.

- **Example** Show that the function $h(x) = \left|x^2 - 3x - 6\right|$ is continuous at each real number.

- **Example** Show that the function $h(x) = \sin \dfrac{x^4 - 3x + 1}{x^2 - x - 6}$ is continuous except at 3 and -2.

- **Intermediate Value Theorem**

> Let f be a function defined on $[a, b]$ and let N be a number between $f(a)$ and $f(b)$. If f is continuous on $[a, b]$, then there is at least a number c between a and b such that $f(c) = N$.



- **Example** Show that the equation $x - \cos x = 0$ has a solution between $x = 0$ and $x = \dfrac{\pi}{2}$.

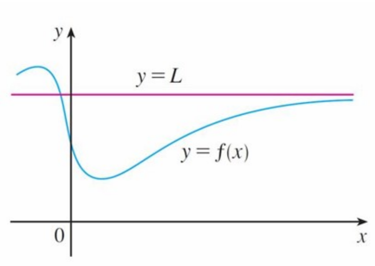
Exercises 2.5 Page 124: 3, 11, 13, 15, 17, 19, 21, 23, 27, 35, 37, 39, 41, 45, 47, 51, 53, 55, 69 (Hint: $x^3 + 1 = x$).

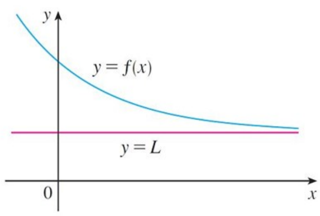
2.6 Limits at Infinity, Horizontal Asymptotes

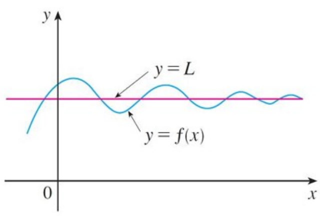
- **Definition**

$$\lim_{x\to\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large.

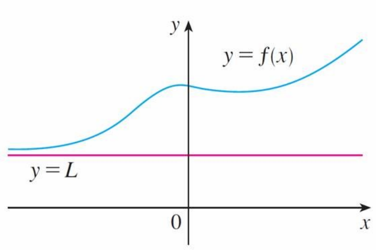


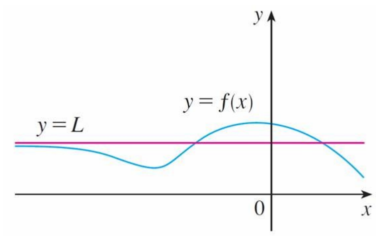




$$\lim_{x\to-\infty} f(x) = L$$

means that the values of $f(x)$ can be made arbitrarily close to L by requiring x to be sufficiently large negative.





- **Example** Find $\displaystyle\lim_{x\to\infty} \frac{x}{1+x^2}$.

- **Example** Find $\lim_{x\to\infty} \frac{2x^3}{1+x^3}$.

- **Example** Find $\lim_{n\to\infty} \sqrt{\frac{n+1}{n+2}}$.

- **Example** Evaluate $\lim_{x\to\infty} \left(\sqrt{x^2+2x}-x\right)$.

- **Example** Find $\lim_{x\to\infty} e^x$ and $\lim_{x\to-\infty} e^x$.

- **Example** Find $\lim_{x\to\infty} \ln x$ and $\lim_{x\to 0^+} \ln x$.

- **Example** Find $\lim_{x\to 0^-} e^{1/x}$

- **Horizontal Asymptotes**

The line $y = L$ is called a **horizontal asymptote** of the curve $y = f(x)$ if either

$$\lim_{x\to\infty} f(x) = L \qquad \text{or} \qquad \lim_{x\to-\infty} f(x) = L$$

Exercises 2.6 Page 137: 3, 13, 15, 17, 19, 21, 23, 25, 27, 29, 31, 33, 35, 37, 39 (Hint: Squeeze Theorem), 41, 47, 49, 51.